



DeepThink Trader

A machine-learning, self-improving paper-trading system that researches, debates, and trades equities with institutional-grade risk discipline.

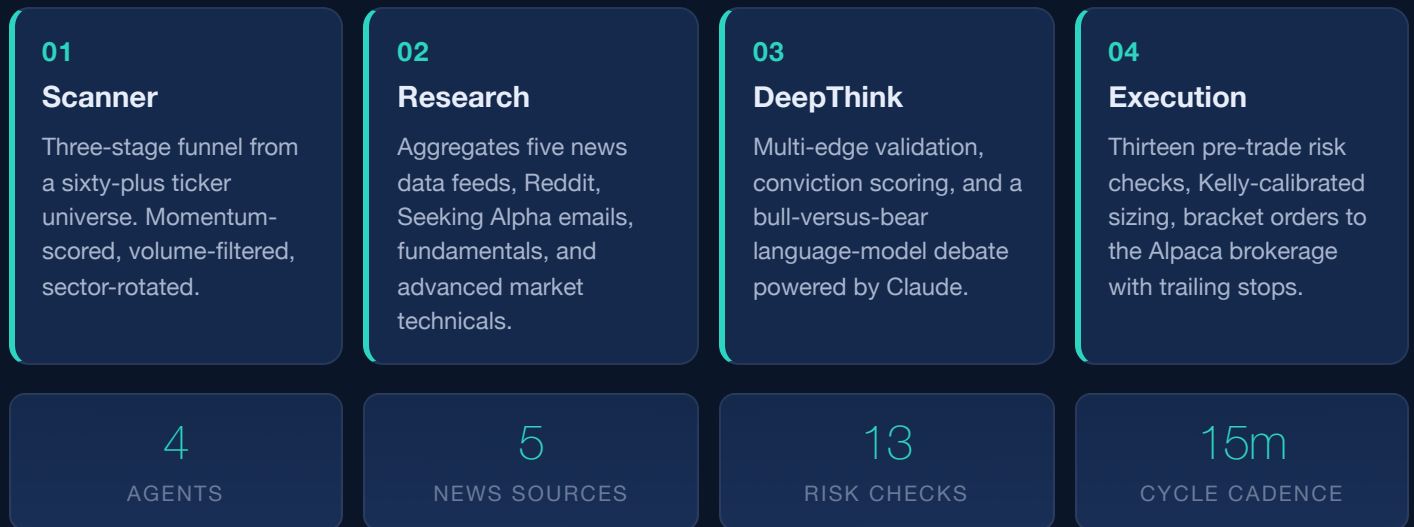


What DeepThinkTrader does

DeepThinkTrader is a fully autonomous trading system that runs continuously during United States market hours. It scans the equity universe, researches candidate tickers from multiple data sources, weighs each through a language-model debate, and places paper trades through a hardened risk gate.

It is built around a four-agent pipeline. Each stage is independently testable, observably logged, and can be paused without bringing the system down.

The pipeline



Operational properties

• Autonomous by default

Boots on system start via the macOS launch agent, runs research cycles every fifteen minutes during market hours, monitors exits every five minutes, refreshes news feeds hourly.

• Fail-safe execution

Every cycle is wrapped in a watchdog that force-restarts on stalls. Startup reconciles ghost positions and stale limit orders against the broker.

• Observable

Heartbeat file, structured log files, Streamlit dashboard. Every language-model call, parameter change, and trade decision is recorded.

• Safety-first

Paper trading only. Strategy auto-pauses if the win rate falls fifteen percent from baseline. A market circuit breaker halts new entries when the S&P 500 index drops two percent in a session.



How the bot determines trades

Every trade decision passes through five sequential checkpoints. A trade enters only when each one passes its threshold; if any fails, the bot holds and the ticker is reconsidered next cycle.

1 Multi-edge validation

Three independent edges — technical (a basket of momentum, trend, volatility, and directional strength indicators), fundamental (earnings momentum, analyst revisions, valuation), and news sentiment (relevance-weighted across five sources). At least two of the three must align.

2 Bull / bear debate

Two language-model personas argue from the same evidence packet for two rounds. A judge model weighs both and emits a winning side, a conviction score from one to ten, and the single most decisive piece of evidence. The losing side's strongest counter becomes the trade's invalidation condition.

3 Conviction gate

Combined score from rule edges and the language-model debate must exceed the active trade mode's threshold (Aggressive 6.0, Normal 7.5, Safe 9.0). Below threshold, the bot holds.

4 Risk gate (thirteen checks)

Kelly-sized position with safety multiplier. Sector exposure cap. Spread guard. Gap risk. Liquidity floor based on average daily volume. Daily loss limit. Drawdown halt. Earnings proximity. Duplicate prevention. Revenge-trade detection. Market circuit breaker. Multi-edge confirmation. Risk-of-ruin probability.

5 Execution and exit management

Bracket order to the Alpaca brokerage with calculated stop-loss and take-profit. Trailing stop activates at two percent profit, trails at one-and-a-half percent. Partial scale-out at one and two risk-units of profit. Time stop at fifteen days. Automatic exit within two days of earnings.

What's in every prompt

• Structured evidence packet

Price action, technical indicators, options flow, fundamentals, earnings calendar, news sentiment per source, Seeking Alpha mention count, Reddit signal.

• Cached for cost

System prompts use the Anthropic ephemeral prompt cache. A twenty-five-minute expiry-based cache reuses ticker analyses if the same ticker is revisited mid-cycle.

Transparent reasoning

Every executed trade carries a plain-English thesis ("why this trade") and an invalidation condition ("what would prove me wrong"). Both are logged to the trade record and surfaced in the dashboard, so every position can be audited after the fact.



Learning from historical activity

DeepThinkTrader is designed to improve over time without manual retuning. Three feedback loops compound: continuous performance measurement, parameter recommendations grounded in trailing metrics, and a captured dataset of every reasoning step the language model has taken.

Three learning loops

● Strategy snapshots

A daily structured record captures the active parameter set alongside trailing thirty-day Sharpe ratio, Sortino ratio, win rate, maximum drawdown, and average return-per-unit-of-risk. Builds the time series the bot uses to detect drift.

Phase 0

Live

● Parameter recommender

A rule engine reads recent snapshots and proposes parameter changes — raising the Kelly fraction when edge is strong, tightening conviction when win rate slips, widening trailing stops when premature exits are observed.

Phase 2

Thirty-day warm-up

● Language-model call dataset

Every Claude prompt and response is logged to a structured file with token usage, latency, and cache-hit metrics. The accumulated corpus is the training set for a future locally-fine-tuned model.

Phase 5 prep

Live

Closed-loop adaptation

Recommendations surface in a daily digest. A human operator reviews them and applies any that look sound. Once a rule has accumulated a track record, it can be flipped to auto-apply via an environment variable. Each automated change is audited; if post-change performance degrades, the change auto-reverts and the rule pauses for thirty days.

Auto-pause and recovery

A weekly health check compares the trailing thirty-day win rate to a rolling ninety-day baseline. If the win rate drops more than fifteen percentage points, the bot halts new entries (exit monitoring continues) until reviewed. When performance recovers, the bot auto-resumes. This prevents a degraded strategy from compounding losses.

Reflection writer

After each closed trade, a structured reflection captures whether the thesis played out, which edges were predictive, and whether the exit reason matched the original invalidation condition. These reflections feed back into the edge-performance tracker that informs future weighting.

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